



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Neoclassical Trade Theory – Standard Trade Model

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Relaxing Assumptions

- Some other new models also relaxing the several assumptions
 - the imitation lag hypothesis (Posner, 1961)
 - the Linder model (Linder, 1961)
 - the flying geese model (Akamatsu, 1961, 1962)
 - the gravity model (Tinbergen, 1962)
 - the product cycle theory (Vernon, 1966)
 - the Krugman model (Krugman, 1979)
 - the reciprocal dumping model (Brander, 1981; Brander and Krugman, 1983)

Dynamic Theory of Comparative Advantage

- dynamic theory of comparative advantage has put greater attention on the changes in supply (production) side
- This is related to how specific determinants affect the output (economic) growth and, in turn, comparative advantage.

- The dynamics of comparative advantage might be also caused by
 - the role of input trade (Jones, 2000)
 - the friction in international trade and investment flows due to geography, institutions, transport, and information cost (Venables, 2001)
 - the transmission of knowledge across borders (Grossman and Helpman, 1991)
 - the technological differences across border (Trefler, 1995)
 - the monopolistic competition in differentiated products with increasing return to scale (Krugman, 1979)

The Production Frontier with Increasing Costs

- **Increasing Opportunity Costs**
 - A nation must give up more and more of one commodity to release just enough resources to produce each additional unit of another commodity.
 - Increasing cost production possibilities frontier is concave to the origin (not a straight line).



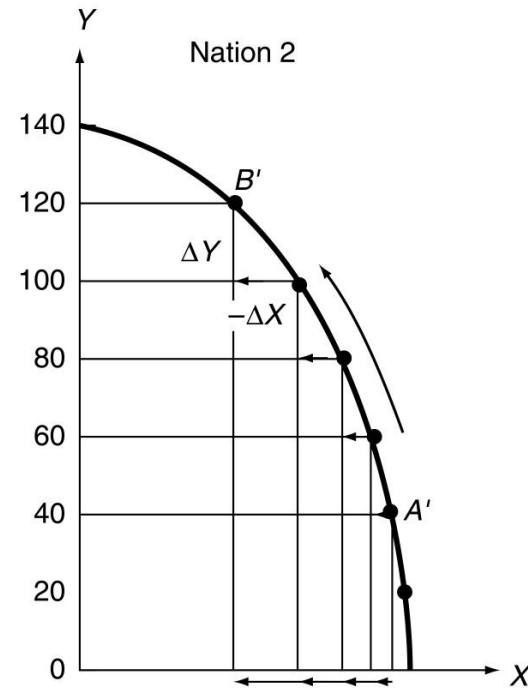
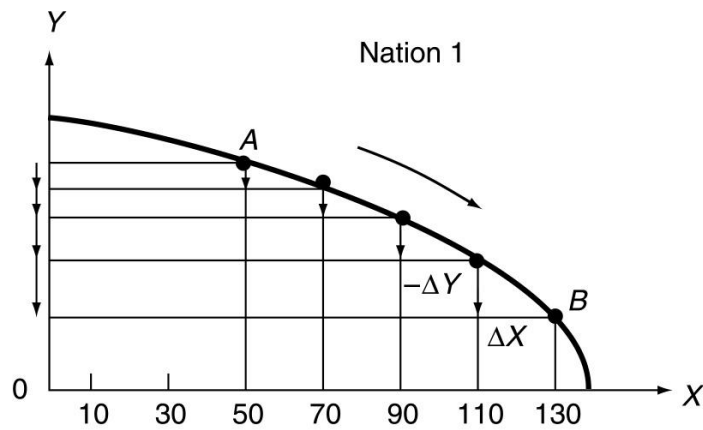
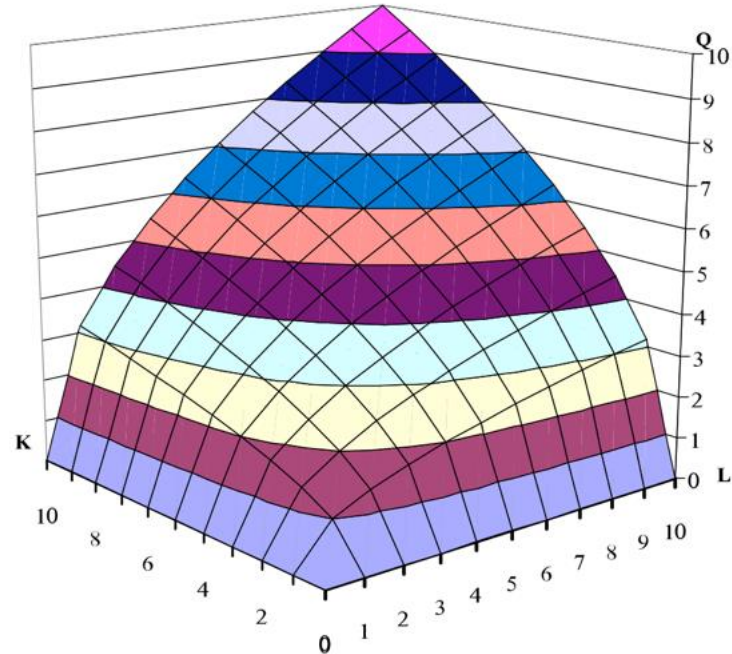


FIGURE Production Frontiers of Nation 1 and Nation 2 with Increasing Costs.

Production Theory

$$X = F(K_X, L_X),$$

$$Y = G(K_Y, L_Y)$$



- Production function

$$X = F(K_X, L_X), \quad Y = G(K_Y, L_Y)$$

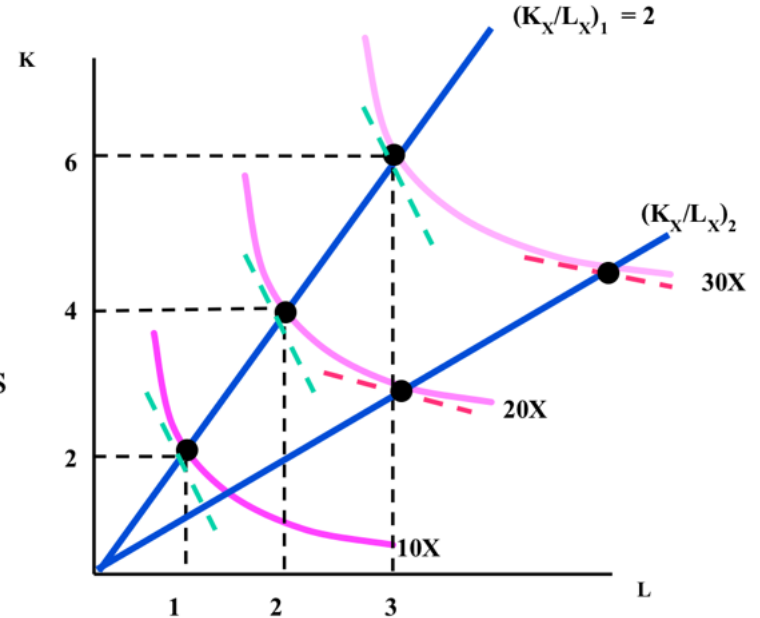
- Constant returns to scale or homogeneous of degree 1

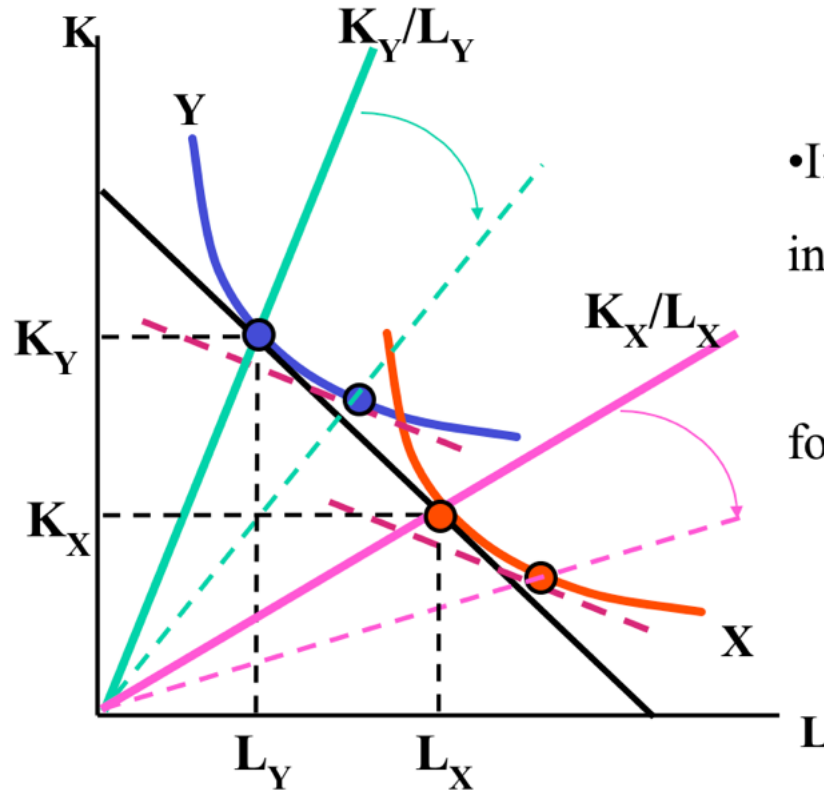
$$_1X = F(_1K_X, _1L_X)$$

$$_1Y = G(_1K_Y, _1L_Y)$$

the output grows at the same rate as the rate of both inputs

- Average products, marginal products, and MRTS are constant along the ray from the origin





- Least cost combination requires

$$MRTS^X = w/r = MRTS^Y$$

- If Y production is capital intensive relative to X,

$$K_Y/L_Y > K_X/L_X$$

for any w/r .

The Production Frontier with Increasing Costs

- The marginal rate of transformation (MRT) increases as more units of one good are produced.
 - The marginal rate of transformation is also called opportunity cost.
 - The value of MRT is given by the slope of the PPF.



Community Indifference Curves

- A **community indifference curve (CIC)** shows combinations of two commodities that derive equal level of satisfaction to the community or nation.
 - Represent measure of taste and preference.
 - Characteristics of community indifference curves



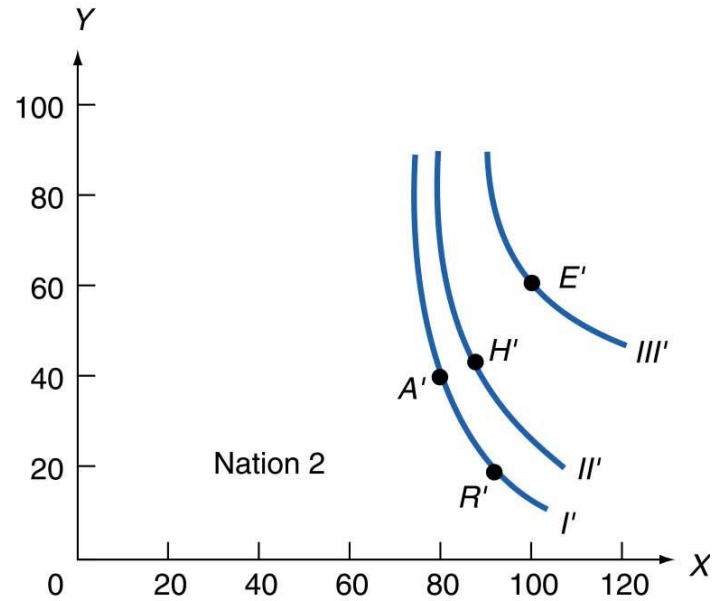
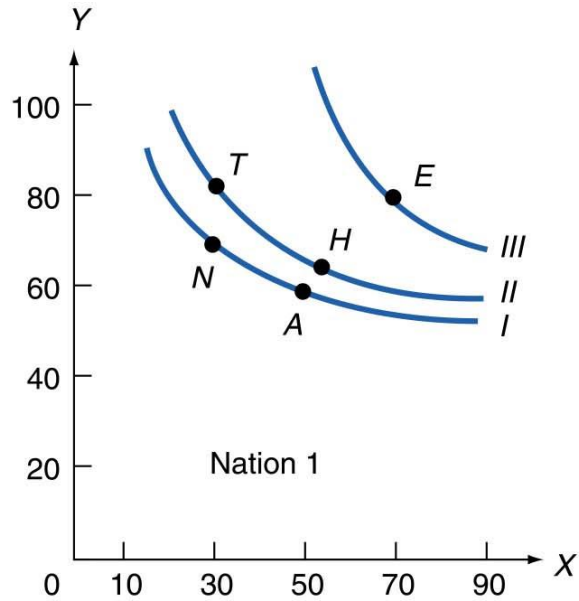


FIGURE Community Indifference Curves for Nation 1 and Nation 2.

Equilibrium in Isolation (Autarky)

- Interaction of forces of demand (CIC) and supply (PPF) determine equilibrium for a nation in the absence of trade (autarky).
- Nations seek the highest possible indifference curve, given its production constraint.



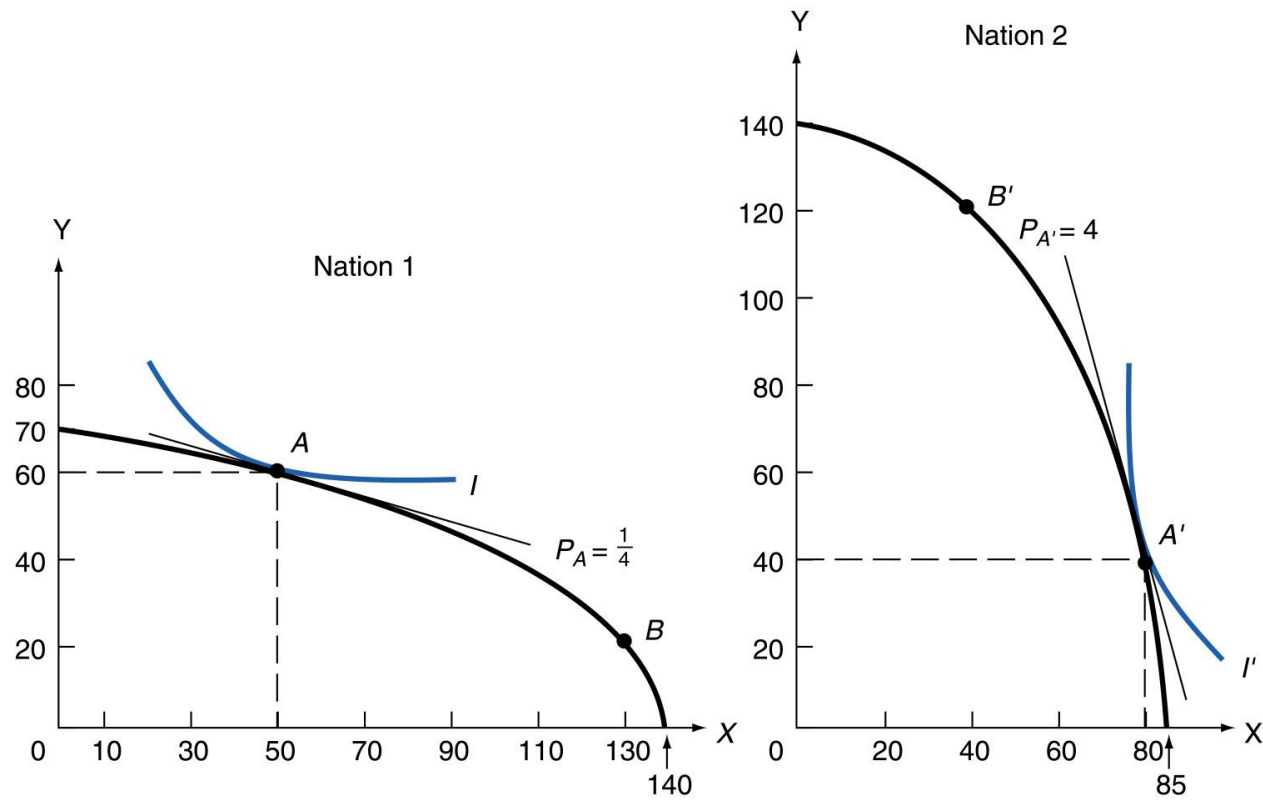
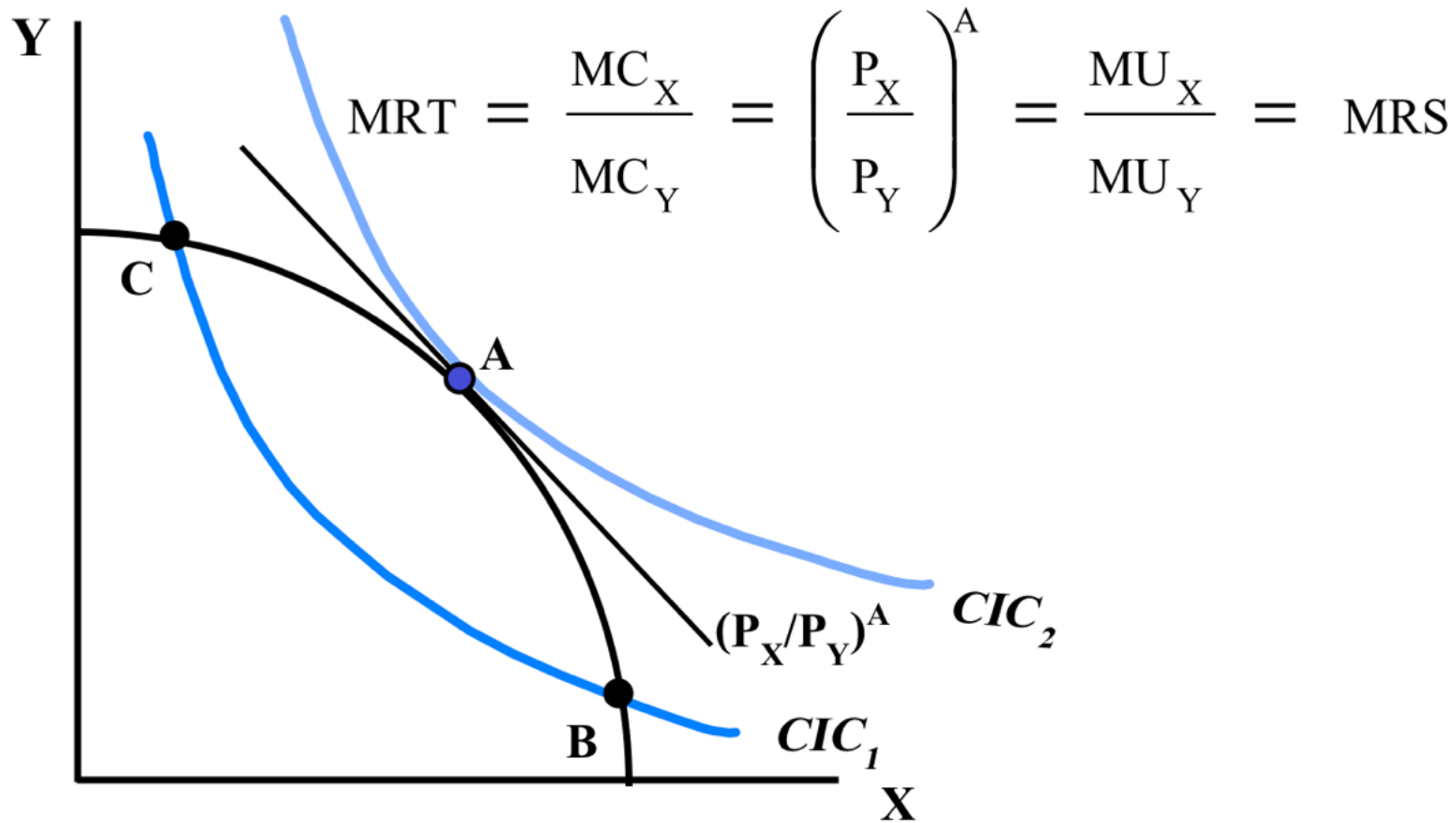


FIGURE 3-3 Equilibrium in Isolation.

Equilibrium in Isolation

- The **equilibrium-relative commodity price** in isolation = slope of tangency between PPF and indifference curve at autarky point of production and consumption.
- Relative prices are different in Nation 1 and Nation 2 because of different shape and location of PPF's and indifference curves.



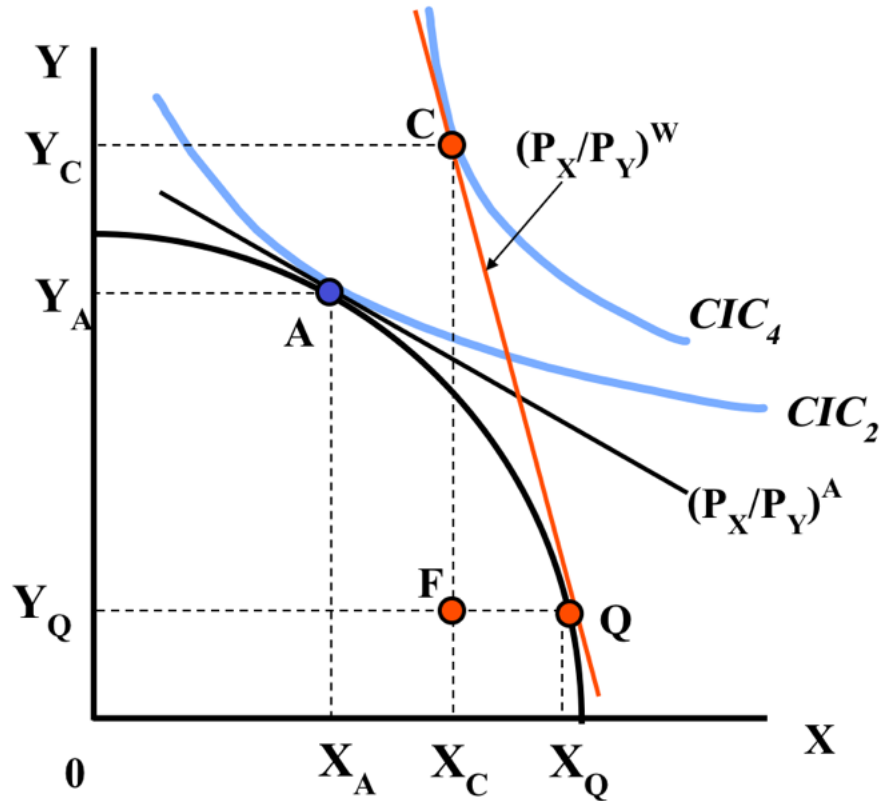


Basis for and Gains from Trade

- Relative commodity price differentials between two nations reflect comparative advantages, and form the basis for mutually beneficial trade.
- Each nation should specialize in the commodity they can produce at the lowest relative price.
- **Specialization will continue until relative prices equalize between nations.**



- Suppose the world relative price of X is higher.



- Production moves to point Q.
- Consumption moves to point C
- Exports X = $X_Q - X_C$.
- Imports Y = $Y_C - Y_Q$.

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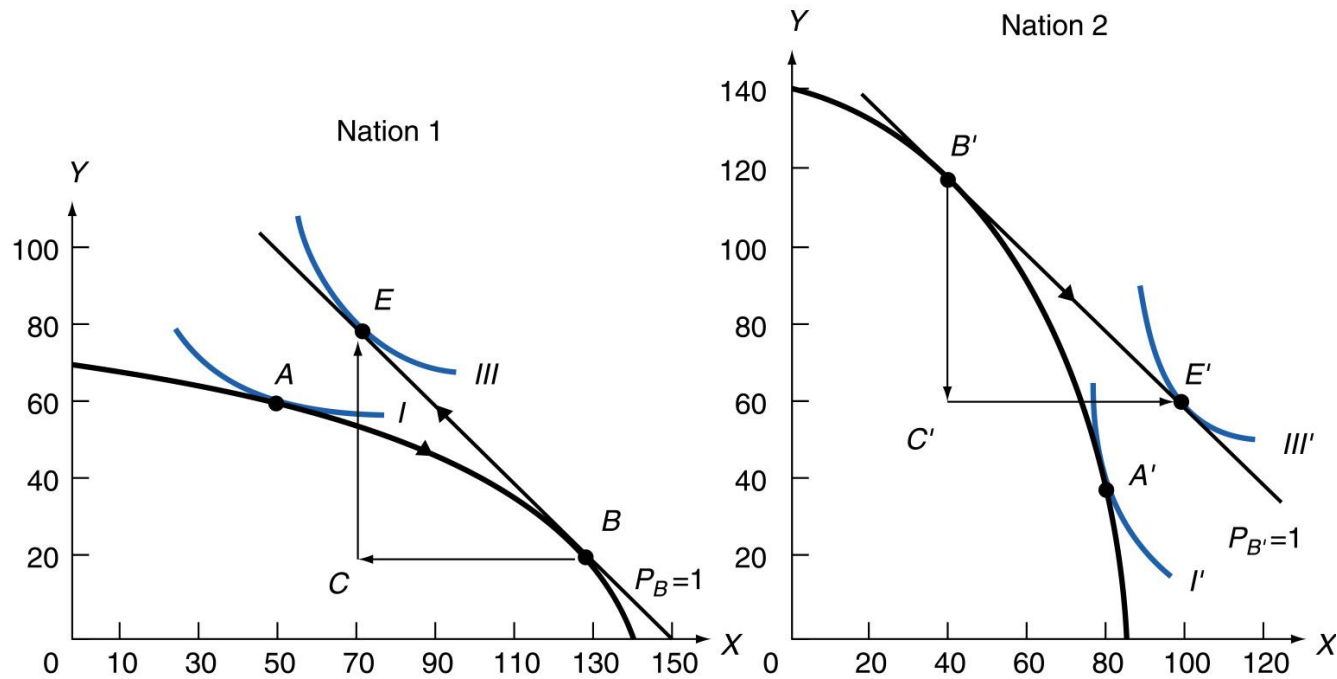


FIGURE The Gains from Trade with Increasing Costs.

BE or B'E' is called *Trade Possibility Line or Trade Line*

Basis for and Gains from Trade

- **Equilibrium-relative commodity price with trade** = common relative price at which trade is balanced.
- **Balanced trade:** quantity of $X_{(Y)}$ Nation 1₍₂₎ wants to export = quantity of $X_{(Y)}$ Nation 2₍₁₎ wants to import.
- Any other relative price could not persist because trade would be unbalanced.



Basis for and Gains from Trade with Increasing Costs

- Under constant cost conditions, specialization is *complete*.
- Under increasing cost conditions, **specialization is *incomplete***:
 - As production moves along PPF toward comparative advantage good, relative costs change, thus changing basis and gains from trade.



Small country with increasing cost

- Under constant cost, complete specialization occurs only in small country case, but large nation continued to produce both commodities even with trade as small country could not satisfy the demand for imports from large nation.
- Still small country would get advantage by exchange at the price ratio and may not specialize, but large country take advantage by exchange as well as specialization.



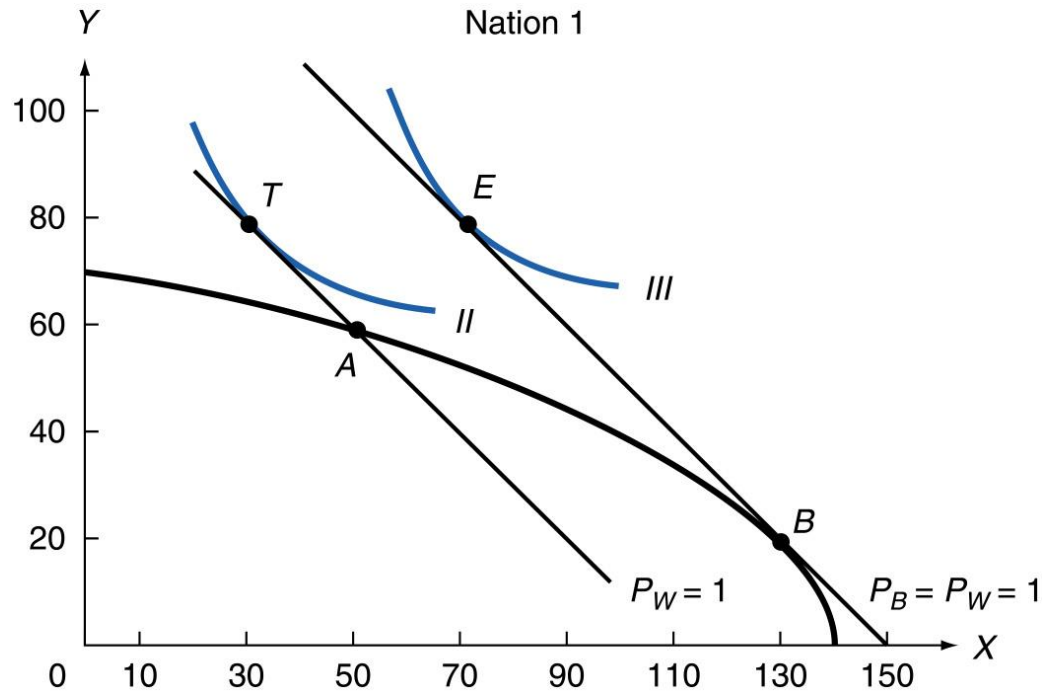


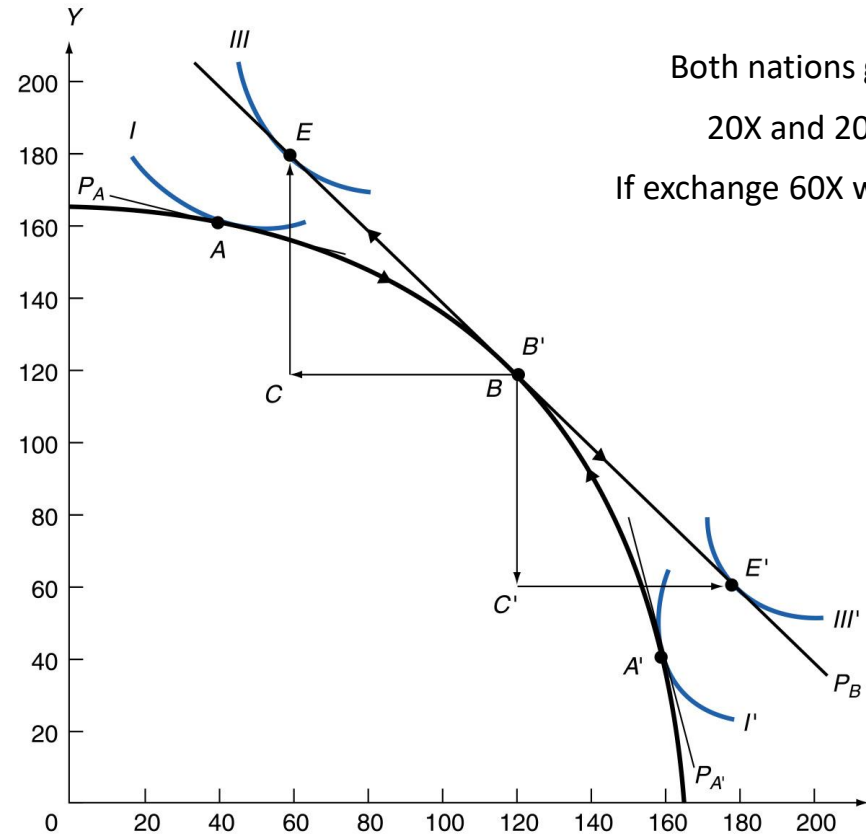
FIGURE The Gains from Exchange and from Specialization.

AT is the possible trade line if country is small.

Trade Based on Differences in Taste

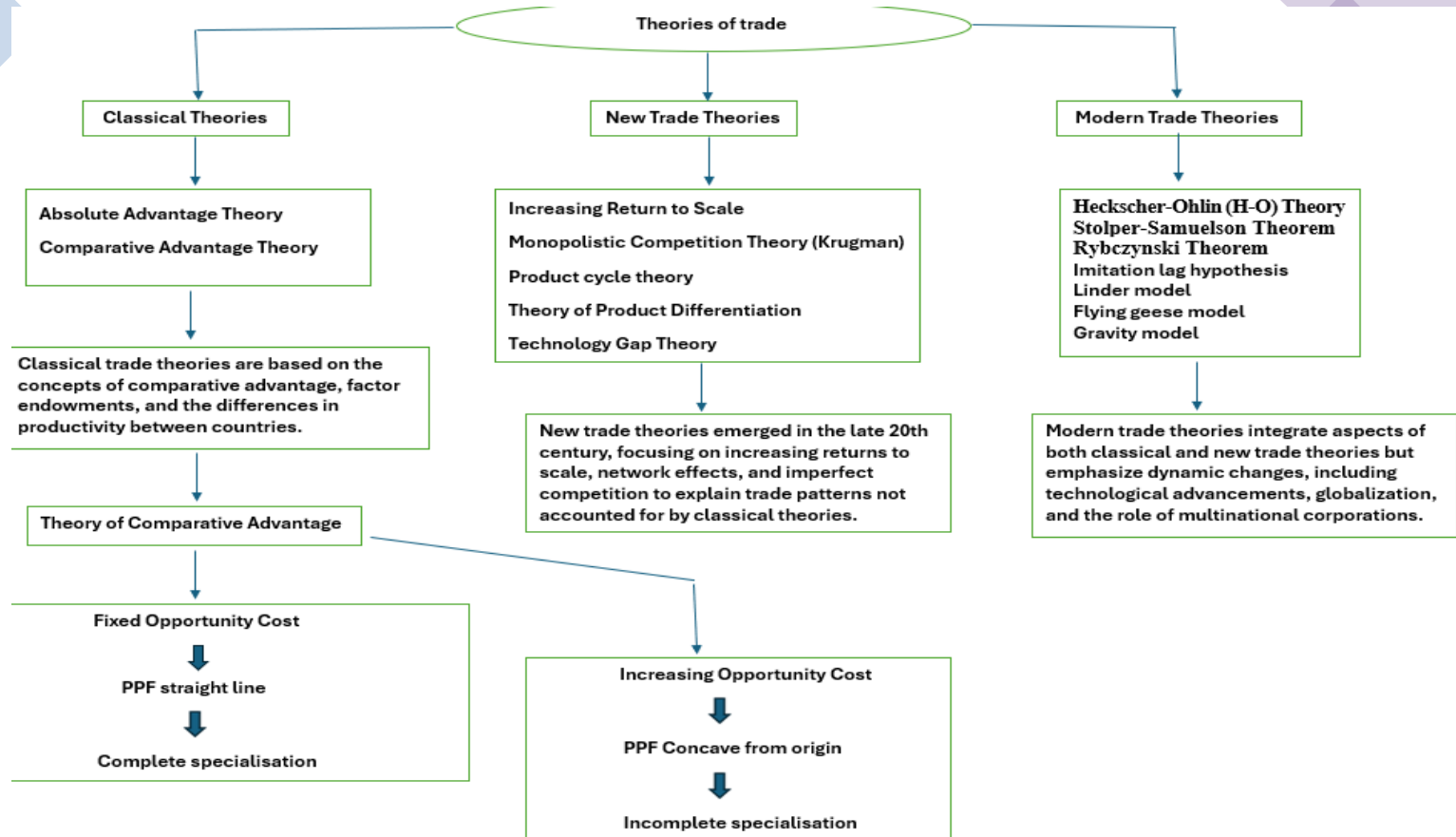
- Even if two nations have identical PPFs, basis for mutually beneficial trade will still exist if tastes, or demand preferences, differ.
- Nation with relatively smaller demand for X will have a lower autarky relative price for, and comparative advantage, in X.





Both nations gain
 20X and 20Y
 If exchange 60X with 60Y

Trade Based on Differences in Tastes.



*Thank
you*

